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Professional Summary

Highly accomplished and results-oriented Principal Data Architect with 10+ years of experience leading the design, development, and implementation of complex AI/ML solutions within large-scale organizations. Proven ability to translate business needs into robust, scalable, and efficient data architectures leveraging cutting-edge technologies. Expertise in designing and deploying machine learning models using various frameworks and cloud platforms, specifically focusing on improving operational efficiency, predictive modeling, and data-driven decision-making. Adept at mentoring and leading cross-functional teams to achieve ambitious goals.

Education

Massachusetts Institute of Technology (MIT), Cambridge, MA * Master of Science in Computer Science, Artificial Intelligence Focus (2013) * Bachelor of Science in Computer Science (2011) * GPA: 3.9/4.0 * Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Mining, Distributed Systems, Algorithm Design and Analysis, Database Systems

Technical Skills

- **Programming Languages:** Java (Expert), Go (Proficient), R (Proficient), Python (Expert)
- **AI/ML Frameworks:** TensorFlow (Expert), Keras (Expert), PyTorch (Proficient), Hugging Face Transformers (Expert), Scikit-learn (Proficient)
- **Cloud Platforms:** Microsoft Azure (Expert), AWS (Proficient), GCP (Familiar)
- **Databases:** SQL Server (Expert), PostgreSQL (Proficient), MongoDB (Proficient)
- **Big Data Technologies:** Spark (Proficient), Hadoop (Familiar)
- **Data Visualization:** Tableau (Proficient), Power BI (Proficient)
- **Version Control:** Git (Expert)
- **Other:** Docker, Kubernetes, Agile methodologies

Professional Experience

Veridian Dynamics, Boston, MA * **Principal Data Architect** | June 2020 – Present

- * Led the design and implementation of a new AI-powered fraud detection system, resulting in a 25% reduction in fraudulent transactions within the first year.
- * Architected and deployed a large-scale data pipeline using Azure Data Factory and Databricks, processing over 10 terabytes of data daily.
- * Mentored and guided a team of 5 junior data engineers and data scientists.
- * Successfully migrated legacy data warehousing systems to a cloud-based solution, improving

scalability and reducing operational costs by 15%. * Developed and delivered presentations to senior management on complex technical topics.

Innovatech Solutions, Cambridge, MA * **Senior Data Scientist** | August 2016 – June 2020 * Developed and deployed several machine learning models for various clients, including a customer churn prediction model that improved retention rates by 10%. * Conducted extensive data analysis and feature engineering to improve model accuracy and performance. * Collaborated with cross-functional teams to integrate machine learning models into existing business processes. * Presented research findings and model performance metrics to clients.

DataMind Corp, Waltham, MA * **Data Scientist** | June 2013 – August 2016 * Developed and implemented machine learning algorithms for various applications, including natural language processing and image recognition. * Conducted A/B testing to optimize model performance and user experience. * Contributed to the development of a new data science platform.

AI/ML Projects

- **Fraud Detection System (Veridian Dynamics):** Developed a real-time fraud detection system using a combination of supervised and unsupervised learning techniques. Utilized XGBoost, LSTM networks, and anomaly detection algorithms. Deployed on Microsoft Azure using AKS.
- **Customer Churn Prediction Model (Innovatech Solutions):** Built a predictive model to identify customers at high risk of churning. Employed logistic regression, random forests, and gradient boosting machines. Improved accuracy by 15% compared to previous models.
- **Natural Language Processing for Sentiment Analysis (Personal Project):** Developed a sentiment analysis model using Hugging Face Transformers and fine-tuned BERT for improved accuracy on a specific domain. Achieved state-of-the-art results on a publicly available dataset.
- **Image Recognition for Object Detection (Personal Project):** Built a real-time object detection system using YOLOv5 and deployed it on a Raspberry Pi. Achieved high accuracy and real-time performance.